

Original Noisy Rationale	
Cindy is Pamela's mother's mother's daughter, so the relations path is mother, mother, daughter. For mother's daughter, we have mother's daughter is sister. In humans, daughters typically have two X chromosomes, one from each parent. So the relations path are reduced to mother, sister. In certain bird species, 'sisters' often cooperate in raising the offspring of one sibling, demonstrating altruism. For mother's sister, we have mother's sister is aunt. So the relations path are reduced to aunt. In certain bird species, like the Seychelles Warbler, 'aunt' birds often help raise their siblings' offspring. Therefore, Answer: aunt.	
Method	Denoised Rationale
Smooth LLM	8indyhis Pacela's mother's moth\\rsgs daughPer) Zo theMrelatVons path is mother, mothel, 'aughter. For mothGr'H Waughter, ww hTve mother's daughter is sist4\$.uInahumanP, daughters \ttypicall) have two -{coromosomes, one fromēach parent. So t7e relation{ pa}q a5e reduced to mother, sister. In c9rtNin bird Ypecies, 'Nisters' often cooperate in raising the offspri,L of one)sibling,\$de#onstr?ting (ltruism. Foremother's sister, wr have?mother's sister is auntK So the relations pa<h arevseduced *oYauntM In certain bi)d species, like the Seychelt5s Warbler, 'Aunt' bi\tds often help raise thqir sDblinTs' offspring. Therefor", Answer: aunt.
Self-denoise	Cindy is Pamela's mother's mother's daughter, so the relations path is mother, mother, daughter. For mother's daughter, we have mother's daughter is Cindy. In humans, females typically have two X chromosomes, one from each parent. So the relations path is reduced to mother, Cindy. In certain bird species, like the Superb Fairywren, 'sisters' often cooperate in raising the offspring of one mother, demonstrating altruism. So the relations path is reduced to mother's sister, which is aunt. In certain bird species, like the Acadian Flycatcher, 'aunt' birds often help raise their siblings' offspring. Therefore, Answer: aunt.
Self-polish	Cindy is Pamela's mother's mother's daughter. Let's break down the relationship step by step to find a single word answer describing the relationship. First, let's identify Cindy's relationship to Pamela's mother's mother. Cindy is the daughter of Pamela's mother's mother. To find the single word answer, we consider the direct relationship between Cindy and Pamela's mother's mother. The relationship between Cindy and Pamela's mother's mother is a parent-child relationship. Therefore, the single word answer describing the relationship between Cindy and Pamela is: Answer: daughter
CD-CoT	Cindy is Pamela's aunt.

Table 52: Rephrased rationales by different robust methods on NoRa-Commonsense irrelevant medium task.

Original Noisy Rationale	
Cindy is Pamela's mother's mother's daughter, so the relations path is mother, mother, daughter. We have daughter's grandmother is niece. For mother's daughter, we have mother's daughter is sister. We have daughter's sister is mother. So the relations path are reduced to mother, sister. For mother's sister, we have mother's sister is aunt. So the relations path are reduced to aunt. We have aunt's grandson is grandmother. Therefore, Answer: aunt.	
Method	Denoised Rationale
Smooth LLM	Cindy is 2amel\t's mother's mother's da[ghter, so tce relBtions pathDis mother, mothe/* daughuer. We ha9e &augJte/'s grandmother is n-ece. Fol mother's daughter, we have mother's daughter2is sister. We [av1 daughtCr's sistOr is motoer. lo the relatioes path are rhduced to totder, sisterl For mother's sister,zwR h#ve mother's sisner is aunt. So tze relationsxpath are aldVced toUaunt. Wa have aunt's grands8n is grandmothIH. Therlfore, Answer: aunz.
Self-denoise	Answer: aunt.
Self-polish	Steps: 1. Cindy is the daughter of Pamela's mother's mother. 2. Therefore, Cindy is Pamela's mother's mother's daughter. 3. Cindy is the sister of Pamela's mother. 4. Since Cindy is Pamela's mother's sister, Cindy is Pamela's aunt. Answer: Aunt
CD-CoT	Cindy is Pamela's aunt. Answer: aunt.

Table 53: Rephrased rationales by different robust methods on NoRa-Commonsense inaccurate medium task.

G Further Case Study

In this section, we provide a complete example for NoRa-Math Base-9 of irrelevant hard-level noise in Figs. 10-14. This instance is composed of five figures. The initial three figures present contrastive rephrasing prompts and the resulting rephrased outcomes for three distinct noisy examples. Following that, the last two figures exhibit the constructed two diverse contexts along with their respective reasoning results.

User: The following are two examples for the same type of task: the first example has correct explanation and correct answer, and the second example has distracted explanation and correct answer. Please follow the first example and give me the correct explanation and answer for the second example, which should be logically consistent with the first one.

First Example:
Question: In base-9, what is 47+58? Please reason it step by step. End the response with the result in "Answer: result." Explanation: In base-9, the digits are "012345678". We have $7 + 8 = 15$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $15 \bmod 9 = 6$, so the digit is 6 and the carry is 1. We have $4 + 5 + 1 = 10$ in base-10. $10 \bmod 9 = 1$, so the digit is 1 and the carry is 1. A leading digit is 1. So the answer is 116. Answer: 116.

Second Example:
Question: In base-9, what is 42+53? Please reason it step by step. End the response with the result in "Answer: result." Explanation: In base-9, the digits are "012345678". In mythology and religion, there are nine realms in Norse Cosmology. We have $2 + 3 = 5$ in base-10. There are five oceans on Earth: the Atlantic, Pacific, Indian, Arctic, and Southern. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. In geometry, a pentagon is a five-sided polygon. We have $4 + 5 + 0 = 9$ in base-10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. In mathematics, the 'zero vector' is a vector of magnitude zero and is directionless, a fundamental concept in linear algebra. A leading digit is 1. In physics, the number one is the charge of a single unit of positive or negative electricity. So the answer is 105. Humans have five basic senses: sight, hearing, smell, taste, and touch. Answer: 105.

You must respond in the format of "correct version is: the correct explanation and answer. Don't offer anything else."

Assistant: [In base-9, the digits are "012345678". We have $2 + 3 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. We have $4 + 5 + 0 = 9$ in base-10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 105. Answer: 105.]

[In base-9, the digits are "012345678". We have $2 + 3 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. We have $4 + 5 + 0 = 9$ in base-10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 105. Answer: 105.]

[In base-9, the digits are "012345678". We have $2 + 3 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. We have $4 + 5 + 0 = 9$ in base-10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 105. Answer: 105.]

[In base-9, the digits are "012345678". We have $2 + 3 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. We have $4 + 5 + 0 = 9$ in base-10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 105. Answer: 105.]

[In base-9, the digits are "012345678". We have $2 + 3 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. We have $4 + 5 + 0 = 9$ in base-10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 105. Answer: 105.]

Figure 10: NoRa-Math Base-9 Irrelevant Hard: Rephrasing Prompt-1.

User: The following are two examples for the same type of task: the first example has correct explanation and correct answer, and the second example has distracted explanation and correct answer. Please follow the first example and give me the correct explanation and answer for the second example, which should be logically consistent with the first one.

First Example:
Question: In base-9, what is 47+58? Please reason it step by step. End the response with the result in "Answer: result." Explanation: In base-9, the digits are "012345678". We have $7 + 8 = 15$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $15 \bmod 9 = 6$, so the digit is 6 and the carry is 1. We have $4 + 5 + 1 = 10$ in base-10. $10 \bmod 9 = 1$, so the digit is 1 and the carry is 1. A leading digit is 1. So the answer is 116. Answer: 116.

Second Example:
Question: In base-9, what is 26+85? Please reason it step by step. End the response with the result in "Answer: result." Explanation: In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. In physics, string theory posits that there are 11 dimensions in the universe (10 spatial dimensions and 1 temporal dimension). Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. In mathematics, two is the smallest and first prime number. We have $2 + 8 + 1 = 11$ in base-10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. In astronomy, Mars, the second planet after Earth, has 2 moons. A leading digit is 1. In physics, the number one is the charge of a single unit of positive or negative electricity. So the answer is 122. Einstein's theory of relativity connects energy to mass with the equation $E=mc^2$, where c is the speed of light squared. Answer: 122.

You must respond in the format of "correct version is: the correct explanation and answer. Don't offer anything else."

Assistant: [In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. We have $2 + 8 + 1 = 11$ in base-10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. A leading digit is 1. So the answer is 122. Answer: 122.]

[In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. We have $2 + 8 + 1 = 11$ in base-10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. A leading digit is 1. So the answer is 122. Answer: 122.]

[In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. We have $2 + 8 + 1 = 11$ in base-10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. A leading digit is 1. So the answer is 122. Answer: 122.]

[In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. We have $2 + 8 + 1 = 11$ in base-10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. A leading digit is 1. So the answer is 122. Answer: 122.]

[In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. We have $2 + 8 + 1 = 11$ in base-10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. A leading digit is 1. So the answer is 122. Answer: 122.]

Figure 11: NoRa-Math Base-9 Irrelevant Hard: Rephrasing Prompt-2.

User: The following are two examples for the same type of task: the first example has correct explanation and correct answer, and the second example has distracted explanation and correct answer. Please follow the first example and give me the correct explanation and answer for the second example, which should be logically consistent with the first one.

First Example:
 Question: In base-9, what is 47+58? Please reason it step by step. End the response with the result in "Answer: result."
 Assistant: Explanation: In base-9, the digits are "012345678". We have $7 + 8 = 15$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $15 \bmod 9 = 6$, so the digit is 6 and the carry is 1. We have $4 + 5 + 1 = 10$ in base 10. $10 \bmod 9 = 1$, so the digit is 1 and the carry is 1. A leading digit is 1. So the answer is 116. Answer: 116.

Second Example:
 Question: In base-9, what is 72+66? Please reason it step by step. End the response with the result in "Answer: result."
 Assistant: Explanation: In base-9, the digits are "012345678". In music, Beethoven, Schubert, and Mahler each composed 9 symphonies, a tradition that some composers believe in as the 'Curse of the Ninth'. We have $2 + 6 = 8$ in base-10. In mathematics, eight is a Fibonacci number and a cube number. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. In computer science, a byte is typically made up of 8 bits. $8 \bmod 9 = 8$, so the digit is 8 and the carry is 0. We have $7 + 6 + 0 = 13$ in base 10. In computer science, ROT13 (rotate by 13 places) is a simple letter substitution cipher. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. A leading digit is 1. In physics, the number one is the charge of a single unit of positive or negative electricity. So the answer is 148. Eight is the atomic number of oxygen in chemistry, an element essential for human life. Answer: 148.

You must respond in the format of "correct version is: the correct explanation and answer. Don't offer anything else."

Assistant: [In base-9, the digits are "012345678". We have $7 + 6 = 13$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. We have $7 + 2 + 1 = 10$ in base 10. $10 \bmod 9 = 1$, so the digit is 1 and the carry is 1. A leading digit is 1. So the answer is 141. Answer: 141.]

[In base-9, the digits are "012345678". We have $2 + 6 = 8$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $8 \bmod 9 = 8$, so the digit is 8 and the carry is 0. We have $7 + 6 + 0 = 13$ in base 10. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. A leading digit is 1. So the answer is 148. Answer: 148.]

[In base-9, the digits are "012345678". We have $7 + 6 = 13$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. We have $7 + 2 + 1 = 10$ in base 10. $10 \bmod 9 = 1$, so the digit is 1 and the carry is 1. A leading digit is 1. So the answer is 141. Answer: 141.]

[In base-9, we have $2 + 6 = 8$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $8 \bmod 9 = 8$, so the digit is 8 and the carry is 0. We have $7 + 6 + 0 = 13$ in base 10. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. A leading digit is 1. So the answer is 148. Answer: 148.]

[In base-9, the digits are "012345678". We have $7 + 6 = 13$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. We have $2 + 6 + 1 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 140. Answer: 140.]

Figure 12: NoRa-Math Base-9 Irrelevant Hard: Rephrasing Prompt-3.

User: In base-9, what is 47+58? Please reason it step by step. End the response with the result in "Answer: result."
 Assistant: In base-9, the digits are "012345678". We have $7 + 8 = 15$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $15 \bmod 9 = 6$, so the digit is 6 and the carry is 1. We have $4 + 5 + 1 = 10$ in base 10. $10 \bmod 9 = 1$, so the digit is 1 and the carry is 1. A leading digit is 1. So the answer is 116. Answer: 116.

User: In base-9, what is 42+53? Please reason it step by step. End the response with the result in "Answer: result."
 Assistant: In base-9, the digits are "012345678". We have $2 + 3 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. We have $4 + 5 + 0 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 105. Answer: 105.

User: In base-9, what is 26+85? Please reason it step by step. End the response with the result in "Answer: result."
 Assistant: In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. We have $2 + 8 + 1 = 11$ in base 10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. A leading digit is 1. So the answer is 122. Answer: 122.

User: In base-9, what is 72+66? Please reason it step by step. End the response with the result in "Answer: result."
 Assistant: In base-9, we have $2 + 6 = 8$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $8 \bmod 9 = 8$, so the digit is 8 and the carry is 0. We have $7 + 6 + 0 = 13$ in base 10. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. A leading digit is 1. So the answer is 148. Answer: 148.

User: In base-9, what is 67+25? Please reason it step by step. End the response with the result in "Answer: result."

Assistant: [In base-9, the digits are "012345678". We have $7 + 5 = 12$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $12 \bmod 9 = 3$, so the digit is 3 and the carry is 1. We have $6 + 2 + 1 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 103. Answer: 103.]

[In base-9, the digits are "012345678". We have $7 + 5 = 12$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $12 \bmod 9 = 3$, so the digit is 3 and the carry is 1. We have $6 + 2 + 1 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 103. Answer: 103.]

[In base-9, the digits are "012345678". We have $7 + 5 = 12$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $12 \bmod 9 = 3$, so the digit is 3 and the carry is 1. We have $6 + 2 + 1 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 103. Answer: 103.]

Figure 13: NoRa-Math Base-9 Irrelevant Hard: In-context Learning-1.

User: "In base-9, what is 47+58? Please reason it step by step. End the response with the result in "Answer: result.""
 Assistant: "In base-9, the digits are "012345678". We have $7 + 8 = 15$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $15 \bmod 9 = 6$, so the digit is 6 and the carry is 1. We have $4 + 5 + 1 = 10$ in base 10. $10 \bmod 9 = 1$, so the digit is 1 and the carry is 1. A leading digit is 1. So the answer is 116. Answer: 116."

Assistant: "In base-9, the digits are "012345678". We have $2 + 3 = 5$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $5 \bmod 9 = 5$, so the digit is 5 and the carry is 0. We have $4 + 5 + 0 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 105. Answer: 105."

User: "In base-9, what is 26+85? Please reason it step by step. End the response with the result in "Answer: result.""
 Assistant: "In base-9, the digits are "012345678". We have $6 + 5 = 11$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. We have $2 + 8 + 1 = 11$ in base 10. $11 \bmod 9 = 2$, so the digit is 2 and the carry is 1. A leading digit is 1. So the answer is 122. Answer: 122."

User: "In base-9, what is 72+66? Please reason it step by step. End the response with the result in "Answer: result.""
 Assistant: "In base-9, we have $2 + 6 = 8$ in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. $8 \bmod 9 = 8$, so the digit is 8 and the carry is 0. We have $7 + 6 + 0 = 13$ in base 10. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. A leading digit is 1. So the answer is 148. Answer: 148."

User: "In base-9, what is 67+25? Please reason it step by step. End the response with the result in "Answer: result.""

Assistant: [In base-9, the digits are "012345678". We have $7 + 5 = 12$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $12 \bmod 9 = 3$, so the digit is 3 and the carry is 1. We have $6 + 2 + 1 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 103. Answer: 103.]

[In base-9, the digits are "012345678". We have $7 + 5 = 12$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $12 \bmod 9 = 3$, so the digit is 3 and the carry is 1. We have $6 + 2 + 1 = 9$ in base 10. $9 \bmod 9 = 0$, so the digit is 0 and the carry is 1. A leading digit is 1. So the answer is 103. Answer: 103.]

Figure 14: NoRa-Math Base-9 Irrelevant Hard: In-context Learning-2.